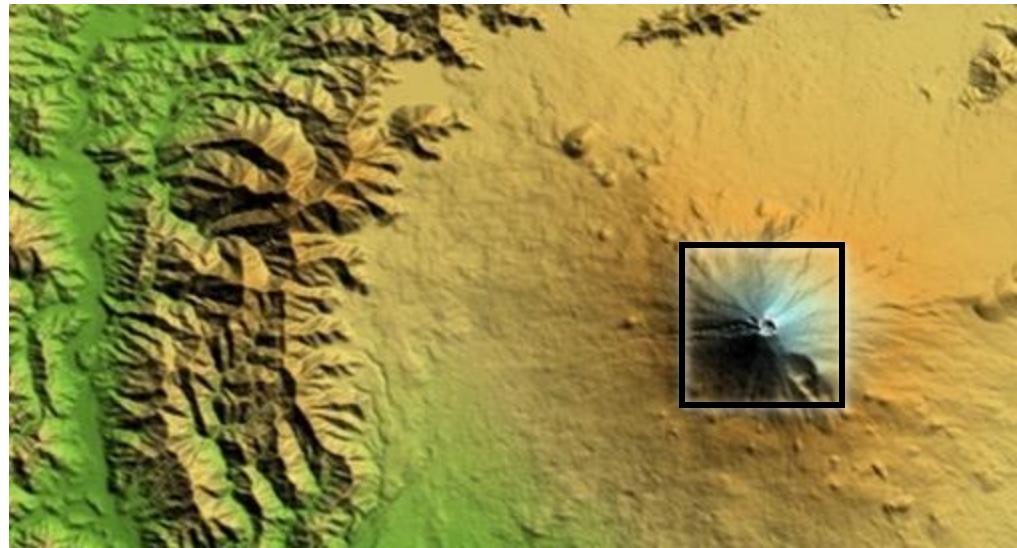


Subgroup Discovery & Exceptional Model Mining

Finding Local Patterns in Data



Local Patterns

- Compared to a *global model*, a *local pattern* is a model that covers only part of the data
- E.g. a prediction rule
 - marital-status = 'Married-civ-spouse' $\wedge$ age $\geq$ 29 $\rightarrow$ target = 'gr50k'
- Statistical relationship only holds for the antecedent (LHS)
 - complement can be same or different
- Multiple local patterns are found, often very many
- Suitable for Exploratory Data Analysis
 - quick insights in attribute dependencies
- Resolve overlap and conflicts
 - especially to create a global model

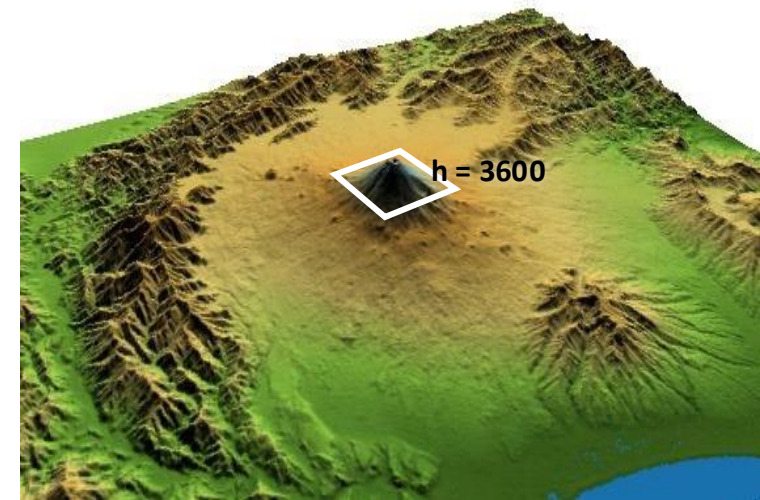
Subgroup Discovery & Exceptional Model Mining

- Subgroup: subset of the data with unusual characteristics
 - Antecedent of a rule
- Subgroup Discovery task

minimum size,
maximum complexity, ...

Find all subgroups within the inductive constraints that show a significant deviation in the distribution of the target attribute

- Subgroup Discovery is a supervised paradigm
 - classification target is binary/nominal
 - regression target is numeric
 - EMM multiple targets
- So supervised, but not a global model



Subgroup Discovery: Binary Target

Confusion Matrix

A *confusion matrix* (or *contingency table*) describes the frequency of the four combinations of subgroup and target:

- within subgroup, positive
 - within subgroup, negative
 - outside subgroup, positive
 - outside subgroup, negative
- } the *complement*

		target		
		T	F	
subgroup	T	0.42	0.13	0.55
	F	0.12	0.33	
		0.54		1.0

Confusion Matrix

- High numbers along the TT-FF diagonal means a *positive* correlation between subgroup and target
- High numbers along the TF-FT diagonal means a *negative* correlation between subgroup and target
- Target distribution on DB is fixed
- Only two degrees of freedom

		target		
		T	F	
subgroup	T	0.42	0.13	0.55
	F	0.12	0.33	0.45
		0.54	0.46	1.0

Quality Measures

A *quality measure* for subgroups summarizes the interestingness of its confusion matrix into a single number

WRAcc, weighted relative accuracy

- $WRAcc(S,T) = p(ST) - p(S) \cdot p(T)$
 - compare $p(ST)$ to independence
- balance between coverage and unexpectedness
- between -0.25 and 0.25 , 0 means uninteresting

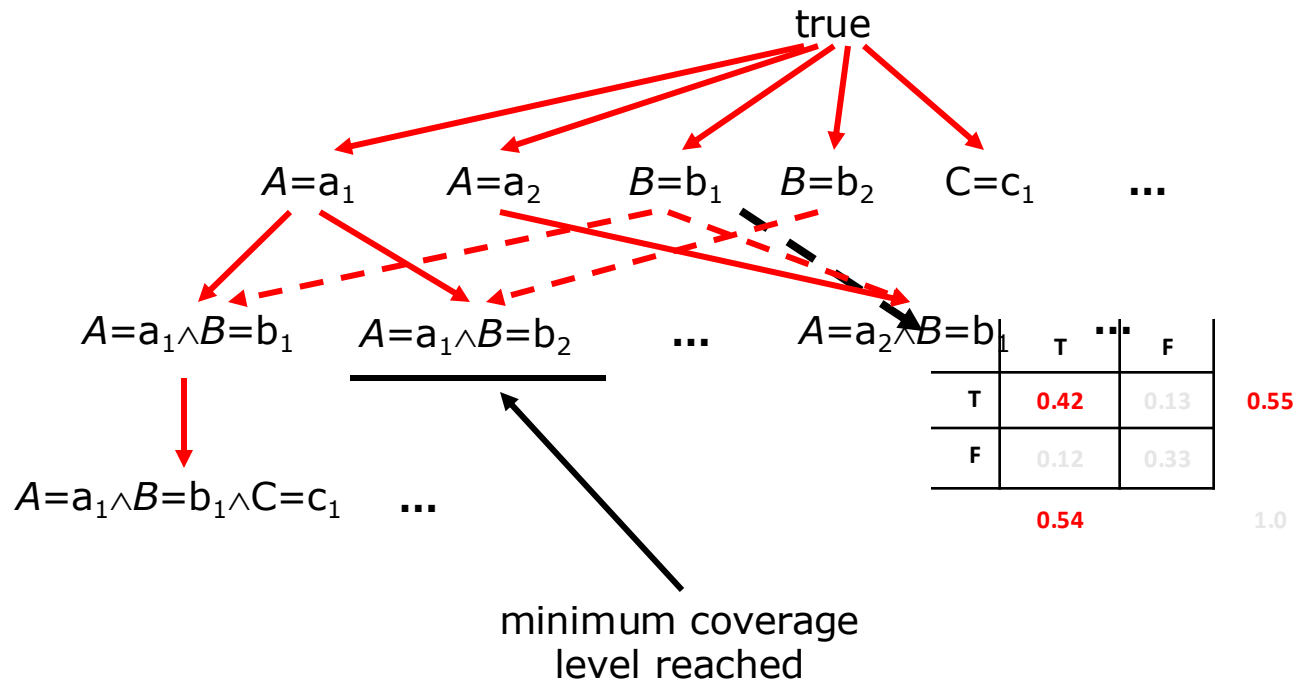
		target		
		T	F	
subgroup	T	0.42	0.13	0.55
	F	0.12	0.33	
		0.54	1.0	

} $WRAcc(S,T) = p(ST) - p(S) \cdot p(T)$
 $= 0.42 - 0.297 = 0.123$

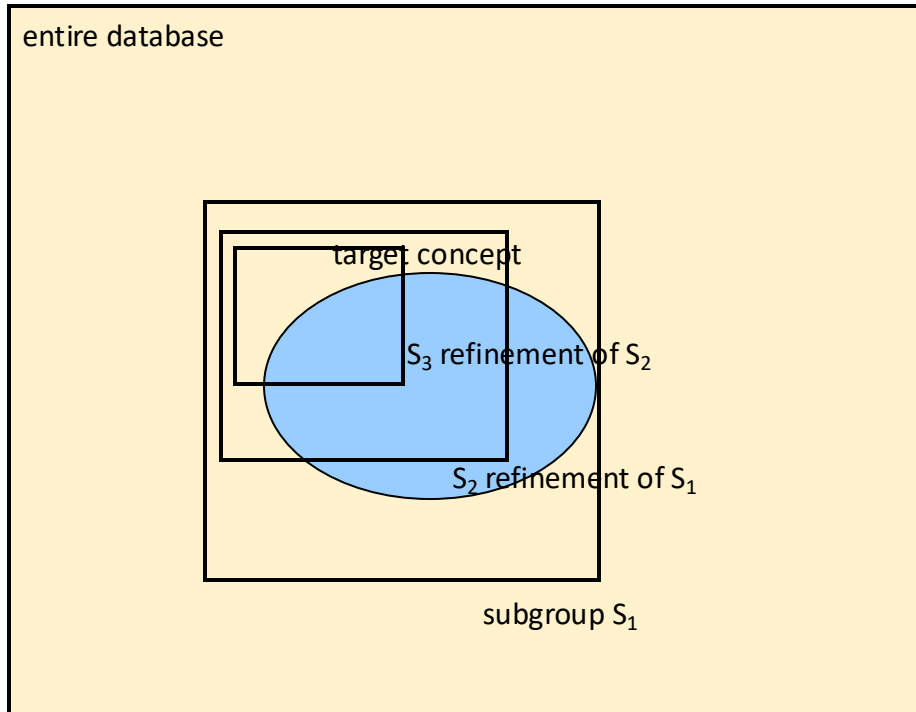
Quality Measures

- WRAcc: Weighted Relative Accuracy
- Cortana Quality
- Information gain
- χ^2
- Correlation Coefficient
- Lift
- Jaccard
- Specificity
- ...

Subgroup Discovery as Search



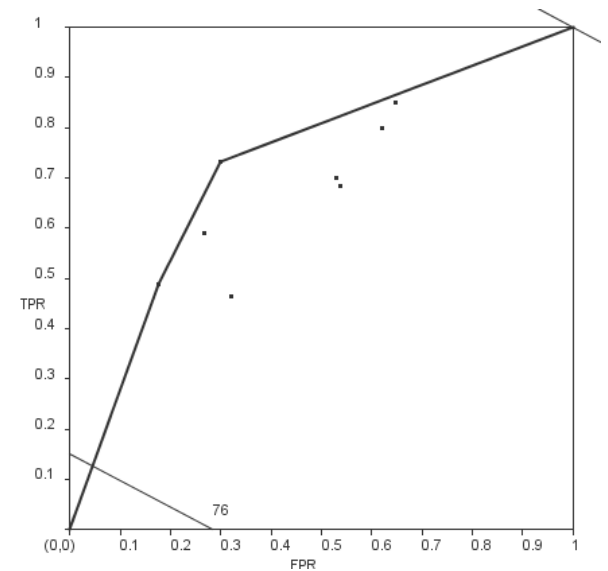
Refinements are (anti-)monotonic



Refinements are (anti-)monotonic in their coverage...

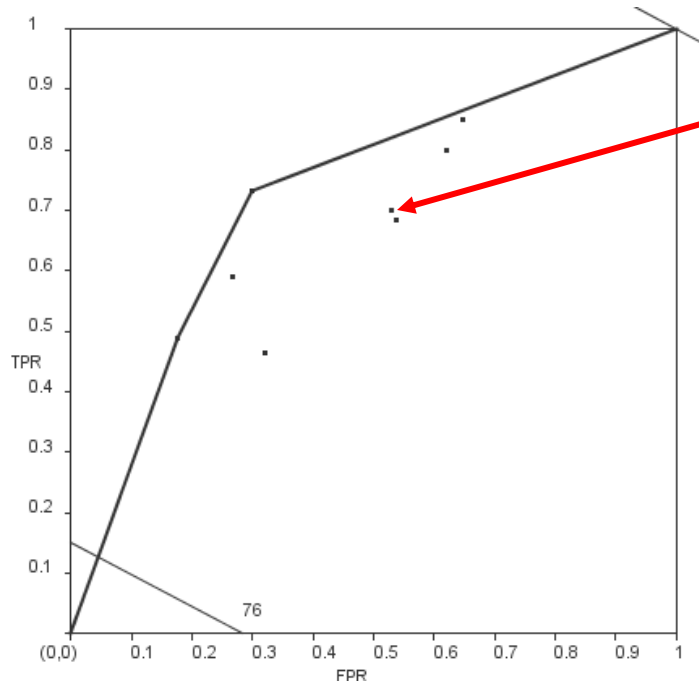
...but not in interestingness. This may go up or down

Subgroup Discovery and ROC space



ROC Space

ROC = Receiver Operating Characteristics



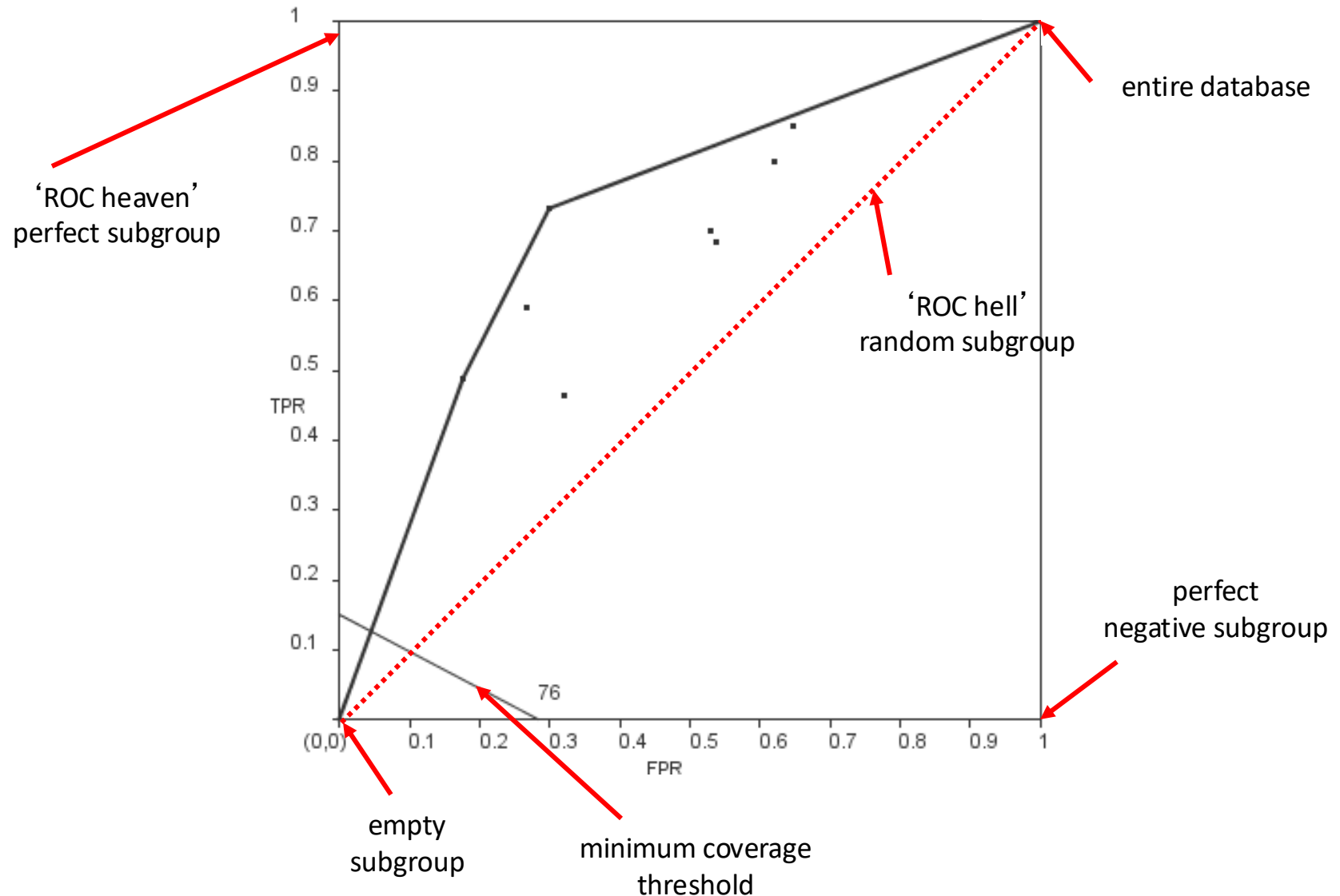
Each subgroup forms a point in ROC space, in terms of its False Positive Rate, and True Positive Rate

indicated subgroup covers about 50% of the negatives and 70% of the positives

$TPR = TP/Pos = TP/(TP+FN)$ (fraction of positive cases in the subgroup)

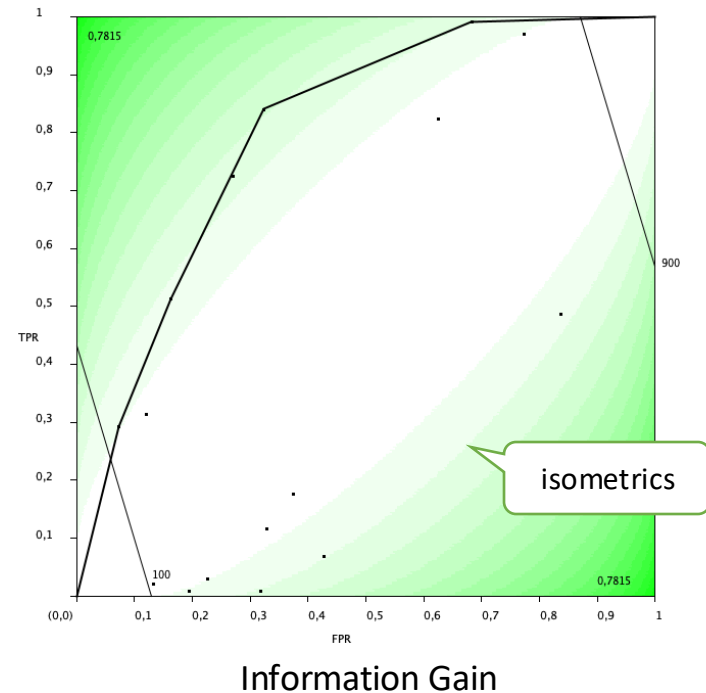
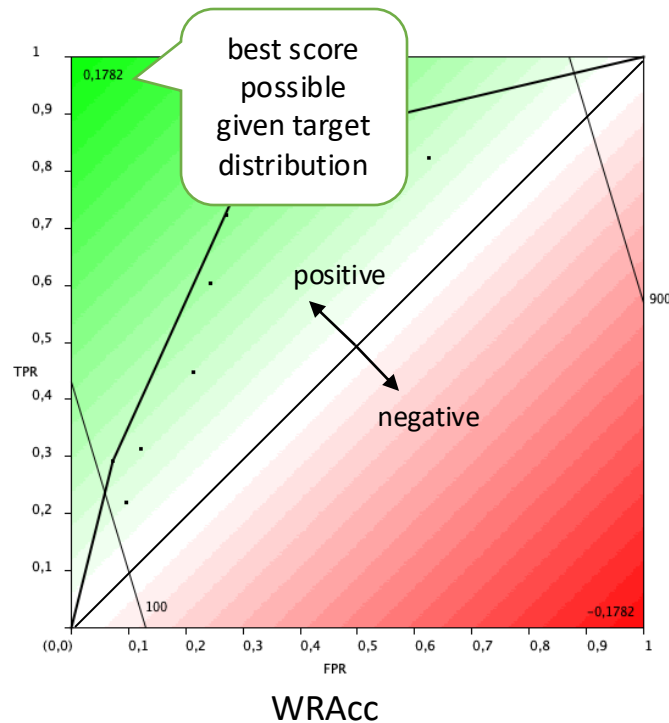
$FPR = FP/Neg = FP/(FP+TN)$ (fraction of negative cases in the subgroup)

ROC Space Properties



Break...

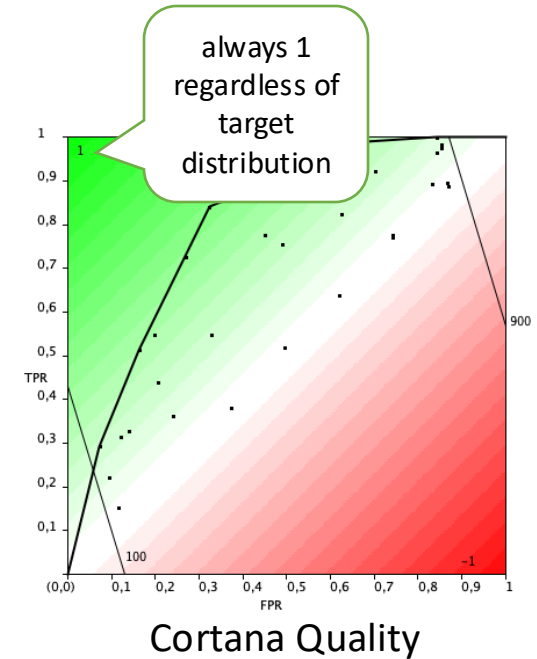
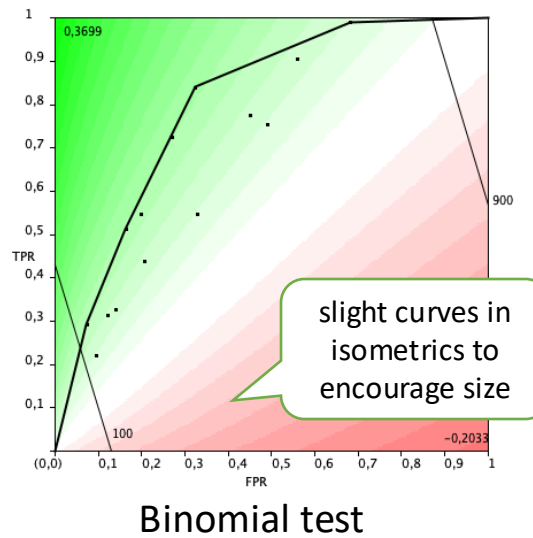
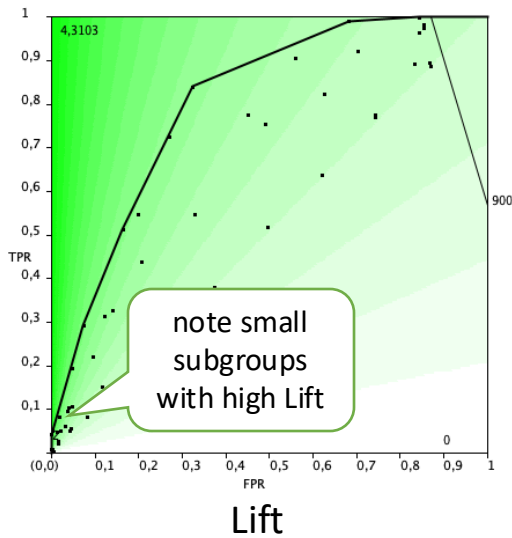
Measures in ROC Space



Properties of WRAcc

- theoretically between -0.25 and 0.25, in practice smaller range (depends on target distribution)
- positive above diagonal, negative below
- isometrics are linear, and parallel to the main diagonal

Behaviour of Common QMs (classification)



Additional properties of Cortana Quality

- between -1 and 1, irrespective of the target distribution
- order-equivalent with WRAcc
 - $\varphi_{cq}(S) = \alpha \cdot \varphi_w(S)$, α is constant per dataset
 - the ranking of subgroups is equivalent for each QM

SubDisc

formerly called Cortana

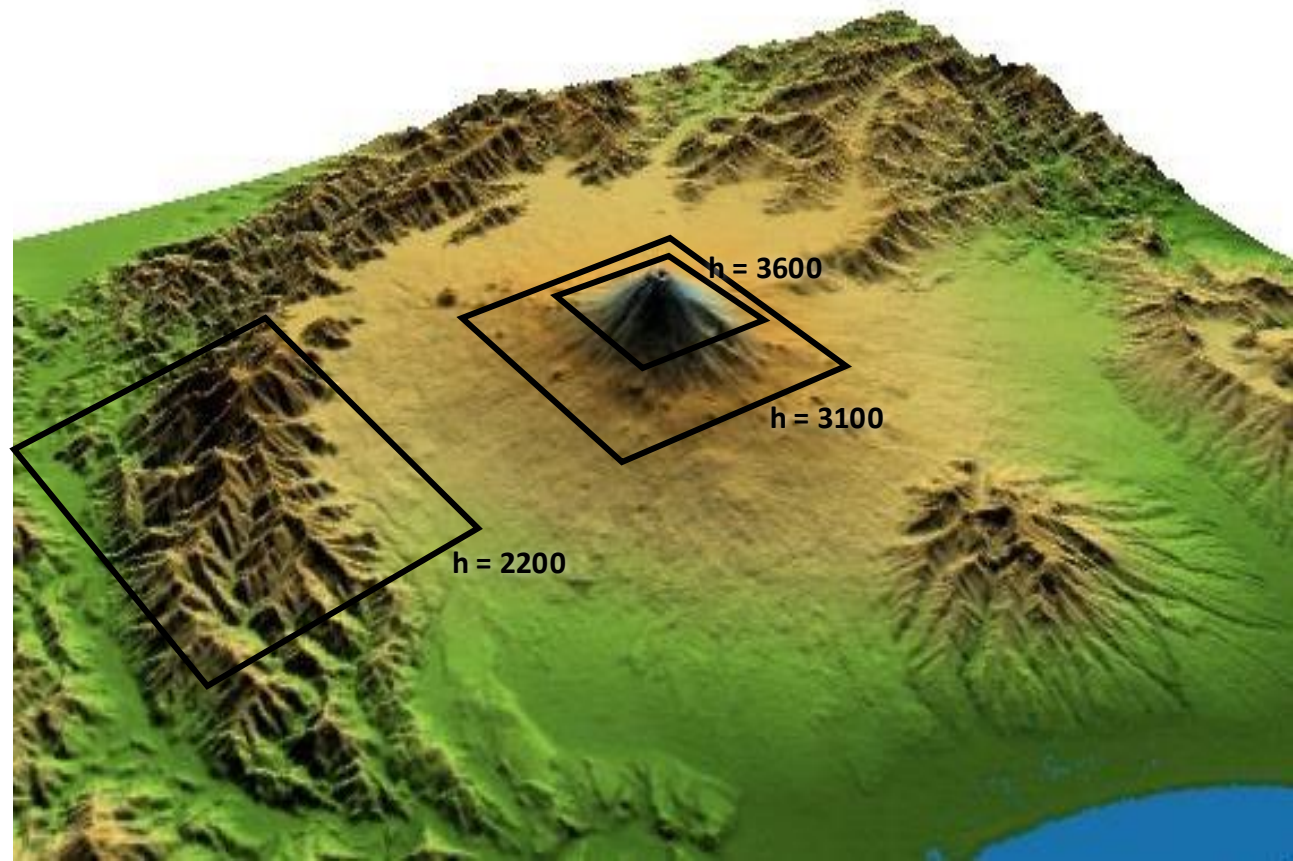
the open source Subgroup Discovery tool
(from Leiden)



Subgroup Discovery for Numeric Targets (regression)

Numeric Subgroup Discovery

- Target is numeric: find subgroups with significantly higher or lower average value
- Trade-off between size of subgroup and average target value





Vancouver 2010 Winter Olympics

	Country	Gold	Silver	Bronze	Total	
1	UNITED STATES	9	15	13	37	
2	GERMANY	10	13	7	30	
3	CANADA	14	7	5	26	
4	NORWAY	9	8	6	23	
5	AUSTRIA		6	6	16	
6	RUSSIAN FEDERATION	3	5	7	15	
7	KOREA	6	6	2	14	
8	CHINA	5	2	4	11	
8	SWEDEN	5	2	4	11	
8	FRANCE	2	3	6	11	

regression target

Official IOC ranking of countries (med > 0)

Rank	Country	Medals	Athletes	Continent	Popul.	Language Family	Repub.	Polar
1	USA	37	214	N. America	309	Germanic	y	y
2	Germany	30	152	Europe	82	Germanic	y	n
3	Canada	26	205	N. America	34	Germanic	n	y
4	Norway	23	100	Europe	4.8	Germanic	n	y
5	Austria	16	79	Europe	8.3	Germanic	y	n
6	Russ. Fed.	15	179	Asia	142	Slavic	y	y
7	Korea	14	46	Asia	73	Altaic	y	n
9	China	11	90	Asia	1338	Sino-Tibetan	y	n
9	Sweden	11	107	Europe	9.3	Germanic	n	y
9	France	11	107	Europe	65	Italic	y	n
11	Switzerland	9	144	Europe	7.8	Germanic	y	n
12	Netherlands	8	34	Europe	16.5	Germanic	n	n
13.5	Czech Rep.	6	92	Europe	10.5	Slavic	y	n
13.5	Poland	6	50	Europe	38	Slavic	y	n
16	Italy	5	110	Europe	60	Italic	y	n
16	Japan	5	94	Asia	127	Japonic	n	n
16	Finland	5	95	Europe	5.3	Finno-Ugric	y	y
20	Australia	3	40	Australia	22	Germanic	y	n
20	Belarus	3	49	Europe	9.6	Slavic	y	n
20	Slovakia	3	73	Europe	5.4	Slavic	y	n
20	Croatia	3	18	Europe	4.5	Slavic	y	n
20	Slovenia	3	49	Europe	2	Slavic	y	n
23	Latvia	2	58	Europe	2.2	Slavic	y	n
25	Great Britain	1	52	Europe	61	Germanic	n	n
25	Estonia	1	30	Europe	1.3	Finno-Ugric	y	n
25	Kazakhstan	1	38	Asia	16	Turkic	y	n

Interesting Subgroups

'polar = yes'

1. United States
3. Canada
4. Norway
6. Russian Federation
9. Sweden
- 16 Finland

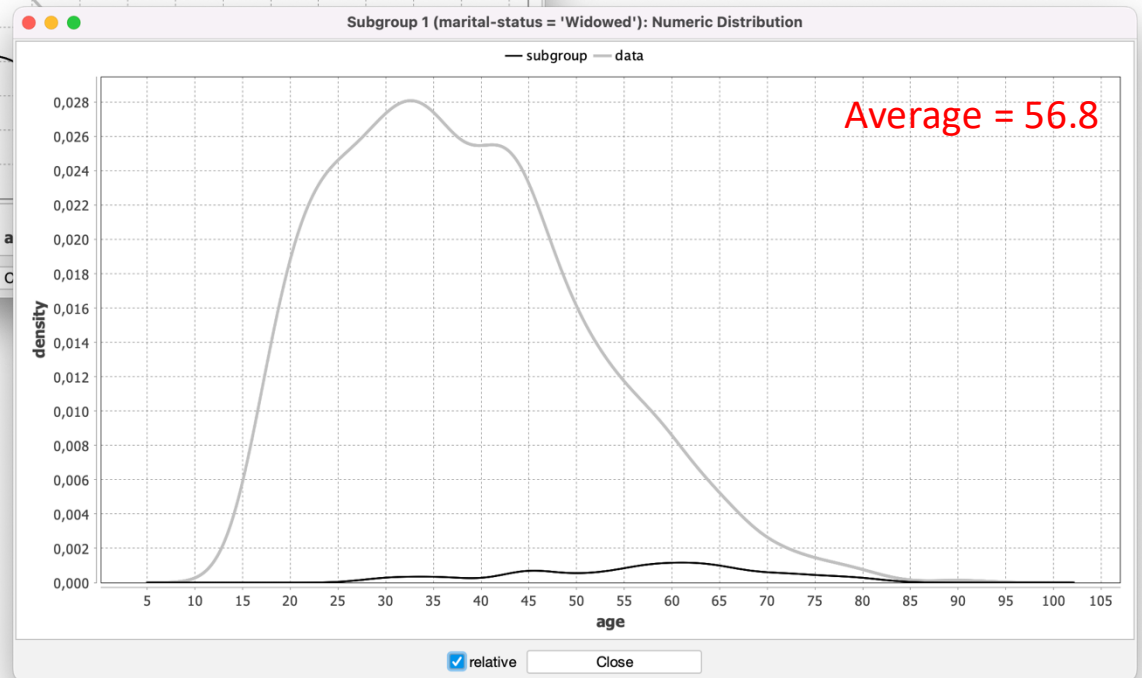
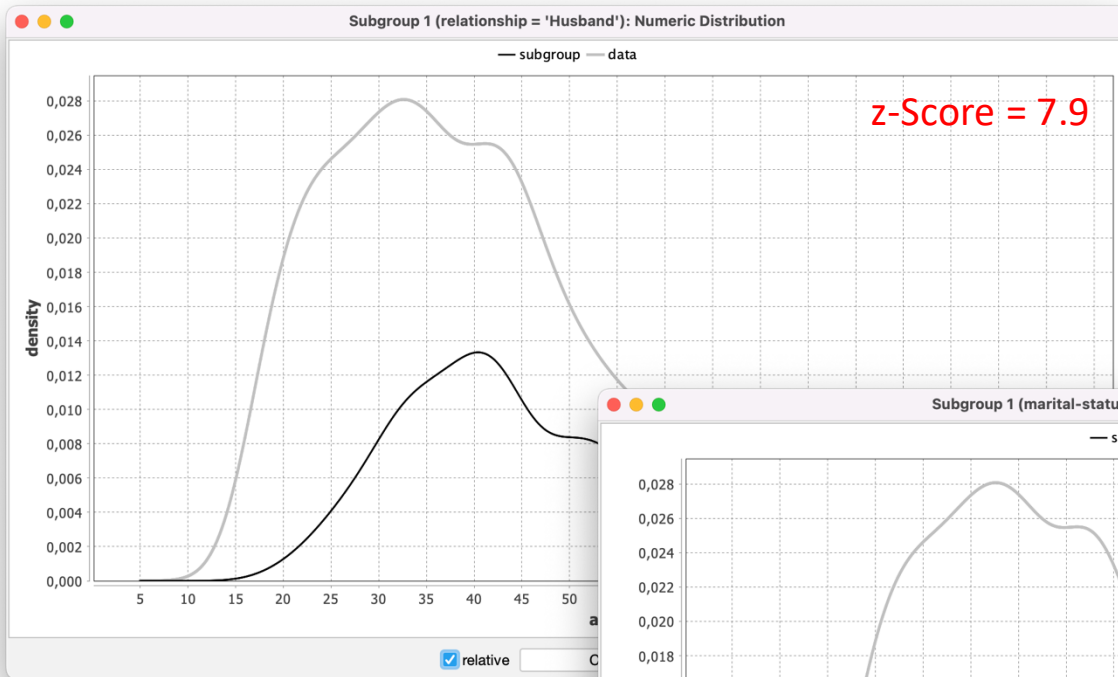
$$n = 6, \mu = 6.5$$

'language_family = Germanic & athletes ≥ 60 '

1. United States
2. Germany
3. Canada
4. Norway
5. Austria
9. Sweden
11. Switzerland

$$n = 7, \mu = 5$$

Best Subgroups for Numeric Target 'Age'



Quality Measures

- **Average**

$$\varphi_{avg}(s) = \frac{\sum_{i=1}^n t_i}{n}$$

- **Mean test**

$$\varphi_{mt}(s) = \sqrt{n}(\mu - \mu_0)$$

- **z-Score**

$$\varphi_z(s) = \frac{\mu - \mu_0}{(\sigma_0 / \sqrt{n})} = \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}$$

- **t-Statistic**

$$\varphi_t(s) = \frac{\mu - \mu_0}{(\sigma / \sqrt{n})}$$

- **Median χ^2 statistic**

$$\varphi_{\chi^2}(s) = \frac{(n_l - N_l)^2}{N_l} + \frac{(n_s - N_s)^2}{N_s}$$

- **AUC of ROC**

$$\varphi_{roc}(s) = \frac{\sum_{i=1}^{\bar{n}} \bar{t}_i - \frac{\bar{n}(\bar{n}+1)}{2}}{\bar{n}n}$$

- **Wilcoxon-Mann-Whitney Ranks statistic**

$$\varphi_{wmw}(s) = \frac{\sum_{i=1}^n t_i - \mu_0}{\sigma_0}$$

- **Median MAD Metric**

$$\varphi_{mmad}(s) = \frac{n}{2 \cdot m + mad}$$

Quality Measures

- **Average**

$$\varphi_{avg}(s) = \frac{\sum_{i=1}^n t_i}{n}$$

- **Mean test**

$$\varphi_{mt}(s) = \sqrt{n}(\dots)$$

Size is not a factor.
(set min cov)

- **z-Score**

order-equivalent

$$\varphi_z(s) =$$

sqrt of size as a factor

$$= \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}$$

- **t-Statistic**

$$\varphi_t(s) = \frac{\mu - \mu_0}{(\sigma / \sqrt{n})}$$

normalise to stdev of target. Makes scores comparable

- **Median X^2 statistic**

$$\varphi_{\chi^2}(s) = \frac{(n_l - N_l)^2}{N_l} + \frac{(n_s - N_s)^2}{N_s}$$

- **AUC of ROC**

$$\varphi_{roc}(s) = \frac{\sum_{i=1}^{\bar{n}} \bar{t}_i - \frac{\bar{n}(\bar{n}+1)}{2}}{\bar{n}n}$$

- **Wilcoxon-Mann-Whitney Ranks statistic**

$$\varphi_{wmw}(s) = \frac{\sum_{i=1}^n t_i - \mu_0}{\sigma_0}$$

- **Median MAD Metric**

$$\varphi_{mmad}(s) = \frac{n}{2 \cdot m + mad}$$

Quality Measures

- Quality measures allow you to specify what you find interesting
- Typically, a trade-off between size and deviation

$$q_S^a(P) = \underbrace{n^a}_{\text{size}} \cdot \underbrace{(t_P - t_0)}_{\text{deviation}}, a \in [0; 1]$$

classification

$$q_S^a(P) = n^a \cdot (t_P - t_0)$$

regression

$$q_M^a(P) = n^a (m_P - m_0),$$

no influence
of size, so set
min cov

$a = 0$

Accuracy, Lift

Average, Abs. Deviation

$a = \frac{1}{2}$

Binomial test

z-Score, Mean test

$a = 1$

WRAcc, Cortana Quality

Impact

SubDisc

formerly called Cortana

the open source Subgroup Discovery tool
(from Leiden)

Target Concepts

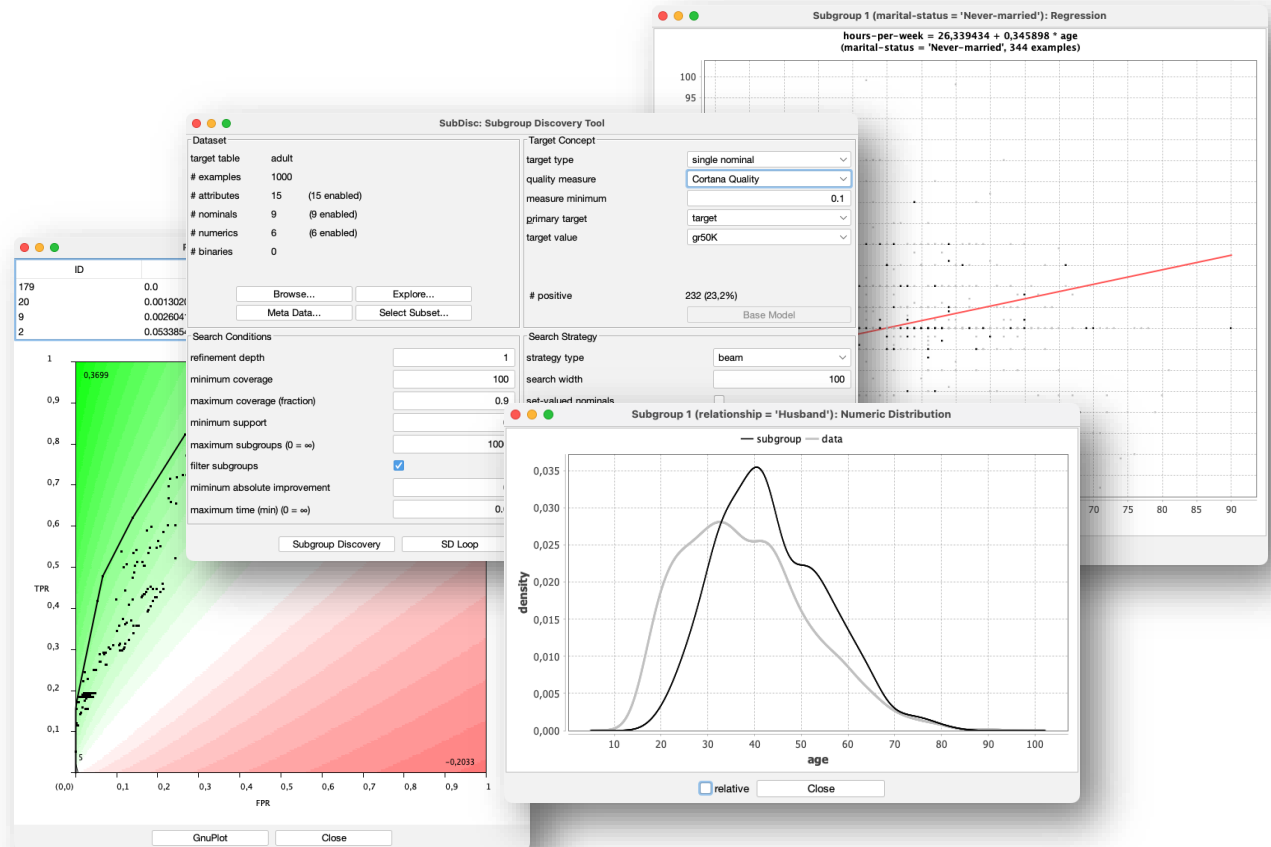
- ‘Classical’ Subgroup Discovery
 - nominal targets (classification)
 - numeric targets (regression)
- Exceptional Model Mining (an extension of SD)
 - multiple targets
 - regression, correlation
 - multi-label classification

Mixed Data

- Data types
 - binary
 - nominal
 - numeric
- Numeric data is treated dynamically (no discretisation as preprocessing)
 - all: consider all available thresholds
 - best: consider all available thresholds, find the best
 - bins: discretise the current refinement attribute
 - best-bins: discretise and continue with best

Open Source

- SubDisc sources and GUI in Java
 - github.com/SubDisc/SubDisc
- SubDisc wrapper in Python and R
 - github.com/SubDisc/pySubDisc
 - github.com/SubDisc/rSubDisc



pySubDisc

- Available from PyPI

```
python3 -m pip install pysubdisc
```

- Example code

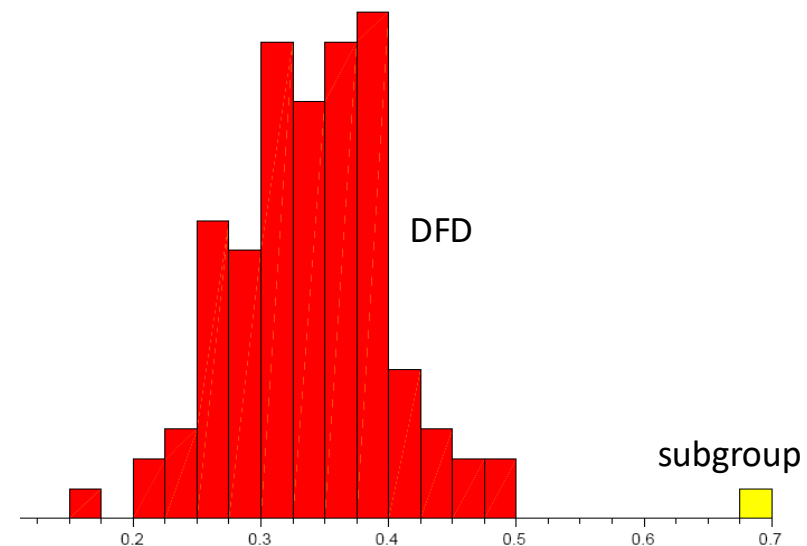
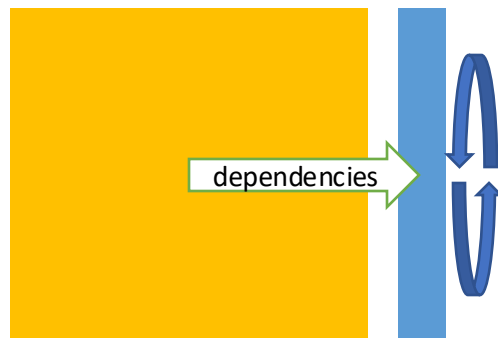
```
import pysubdisc  
import pandas
```

```
data = pandas.read_csv('adult.txt')  
sd = pysubdisc.singleNominalTarget(data, 'target', 'gr50K')  
sd.qualityMeasureMinimum = 0.25  
sd.run()  
print(sd.asDataFrame())
```

Break

Swap Randomisation

- Subgroup Discovery does extensive search
 - risk of finding spurious dependencies
- Mitigate this risk by comparing against similar runs with permuted target
- Swap randomisation of target
 - all other data remains intact
 - distribution of target remains intact
 - dependencies between predictors and target removed
- Compare with same search settings!



Statistical Validation in pySubDisc

- Set the minimum quality threshold before running
- This requires that all other parameters are set before

...

```
#run 100 swap-randomised SD runs
```

```
sd.computeThreshold(significanceLevel=0.05, setAsMinimum=True)
```

```
print("Minimum quality for significance: ", sd.qualityMeasureMinimum)
```

```
sd.run()
```

...

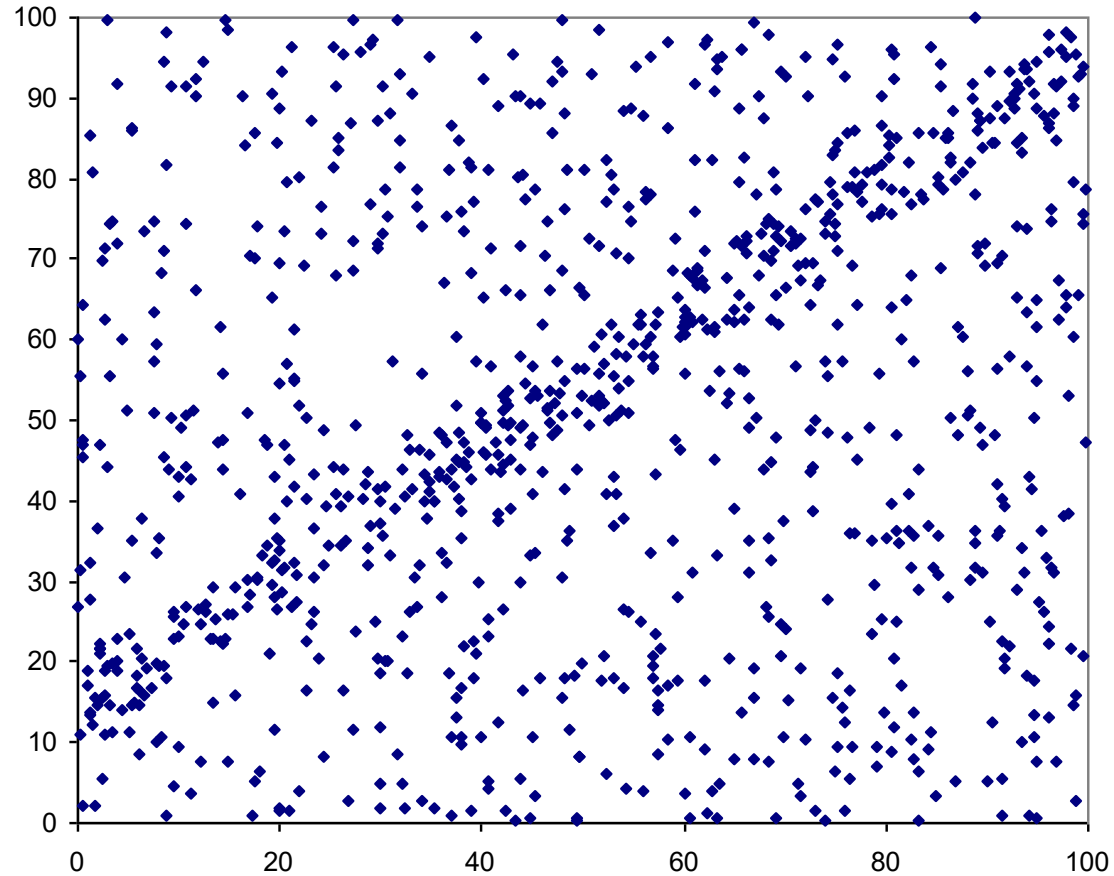
Pattern Explosion

- Easy to generate very many subgroups
- Often, slight variations on a theme
 - marital-status = 'Married-civ-spouse' AND age >= 29 AND education-num >= 8
 - marital-status = 'Married-civ-spouse' AND age >= 29 AND hours-per-week >= 24
 - marital-status = 'Married-civ-spouse' AND age >= 29 AND fnlwgt <= 445382
 - marital-status = 'Married-civ-spouse' AND age >= 29 AND age <= 76
 - age <= 67 AND marital-status = 'Married-civ-spouse' AND age >= 29
- Generalisation-aware subgroup discovery
 - Refine a subgroup, get a worse score
 - 0.518, marital-status = 'Married-civ-spouse'
 - 0.461, marital-status = 'Married-civ-spouse' AND sex = 'Male'
 - Marginal gains
 - 0.556, marital-status = 'Married-civ-spouse' AND age >= 29
 - 0.560, marital-status = 'Married-civ-spouse' AND age >= 29 AND age <= 76
- Pattern Teams
 - Select a subset of subgroups that optimises the joint entropy of the 'extension' (membership bitvector)

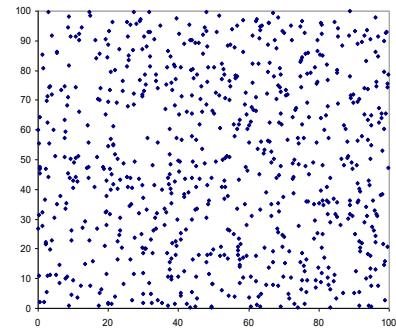
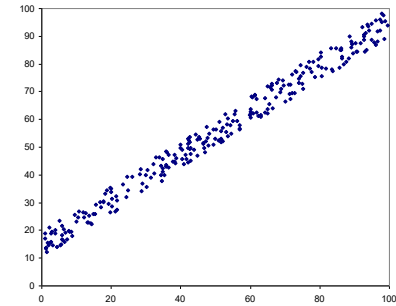
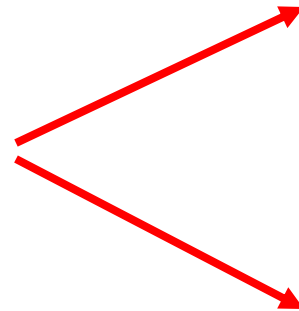
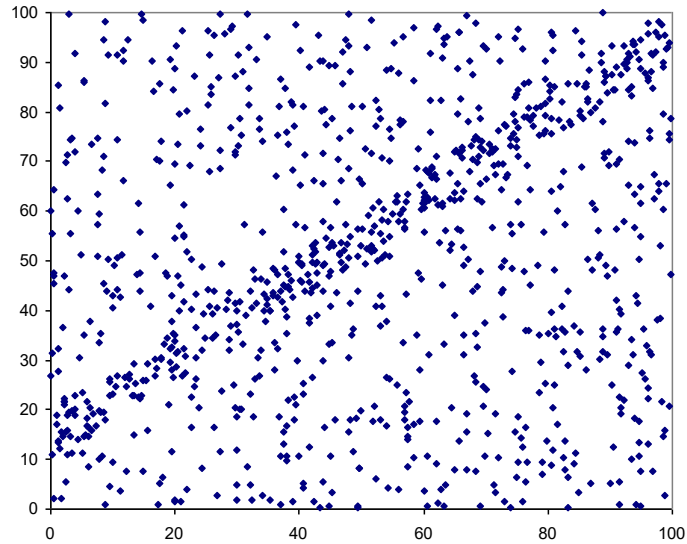
Exceptional Model Mining

Subgroup Discovery with multiple target attributes

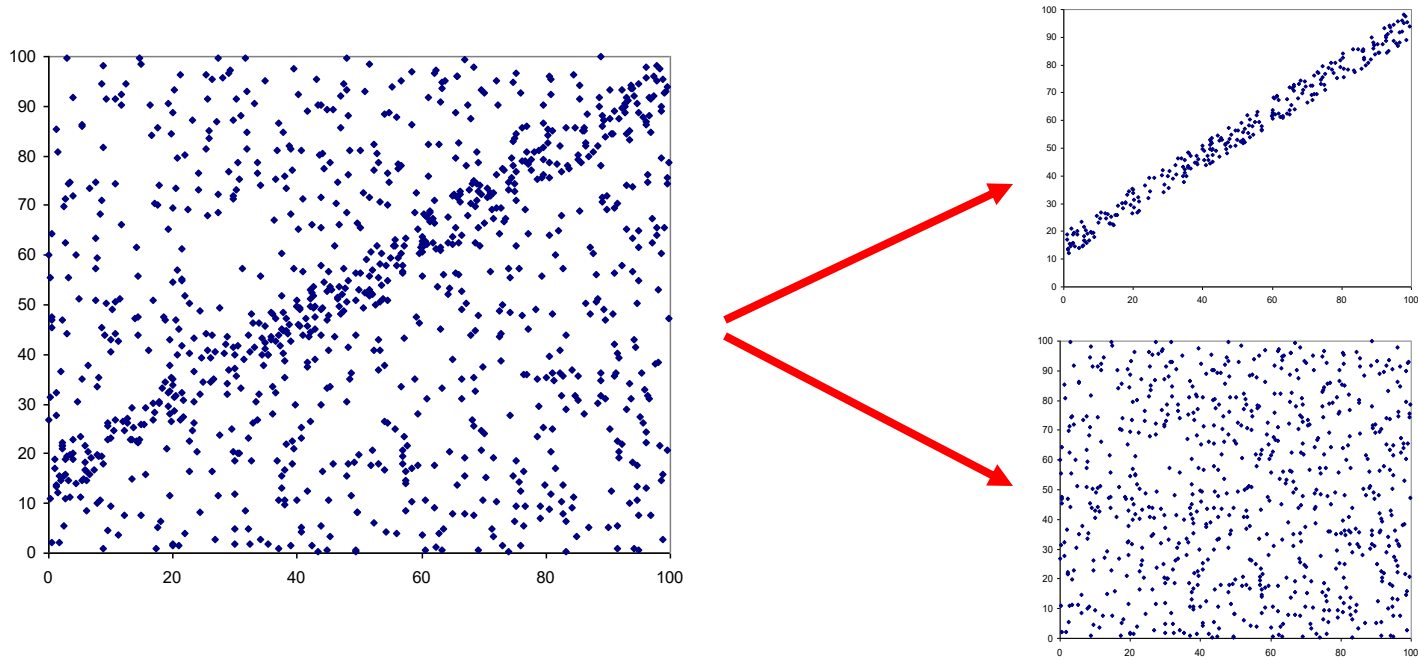
Mixture of Distributions



Mixture of Distributions



Mixture of Distributions



- For each data point, it is unclear whether it belongs to S or $\bar{S}$
- Intensional description of exceptional subgroup S ?
- Model parameters unknown

Solution: Extend Subgroup Discovery

- Use other information than x and y : object descriptions S
- Use Subgroup Discovery to scan sub-populations in terms of S

Subgroup Discovery: find subgroups of the database where the target attribute shows an unusual distribution

Solution: Extend Subgroup Discovery

- Use other information than x and y : object descriptions S
- Use Subgroup Discovery to scan sub-populations in terms of S
- Model over subgroup becomes target of SD

Exceptional Model Mining

~~Subgroup Discovery~~: find subgroups of the database where the target attributes show an unusual distribution, *by means of modeling over the target attributes*

Exceptional Model Mining



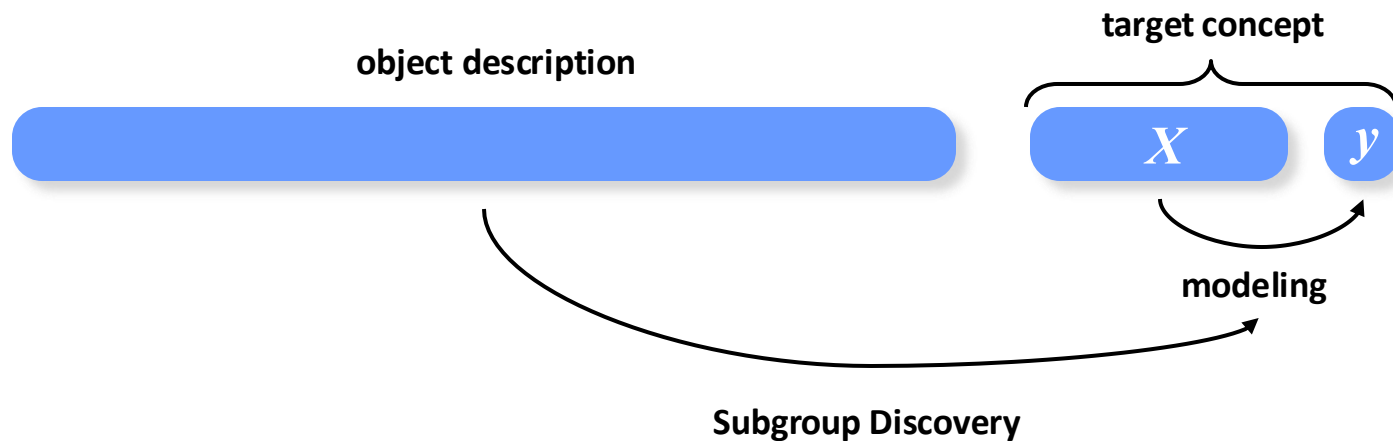
- Define a target concept (X and y)

Exceptional Model Mining



- Define a target concept (X and y)
- Choose a model class C
- Define a quality measure φ over C

Exceptional Model Mining



- Define a target concept (X and y)
- Choose a model class C
- Define a quality measure φ over C
- Use Subgroup Discovery to find exceptional subgroups S and associated model M

Quality Measure

- Specify what defines an exceptional subgroup S based on properties of model M
- Absolute measure (absolute quality of M)
 - Correlation coefficient
 - Predictive accuracy
- Difference measure (difference between M and $\overline{M}$)
 - Difference in slope
 - Qualitative properties of classifier
- Reliable results
 - Minimum support level
 - Statistical significance of S

Correlation Model

- Correlation coefficient

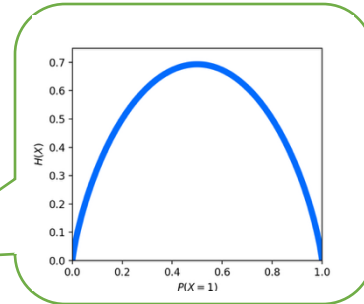
$$\varphi_{\rho} = \rho(S)$$

- Absolute difference in correlation

$$\varphi_{abs} = |\rho(S) - \rho(\bar{S})|$$

- Entropy weighted absolute difference

$$\varphi_{ent} = H(p) \cdot |\rho(S) - \rho(\bar{S})|$$



- Statistical significance of correlation difference φ_{scd}
 - compute z-score from ρ through Fisher transformation

$$z' = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right)$$

- compute p -value from z-score

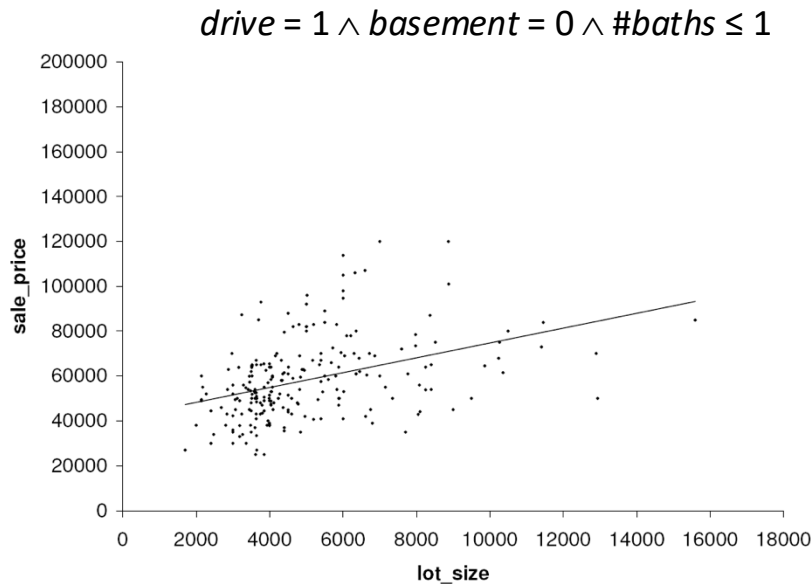
Regression Model

- Compare slope b of

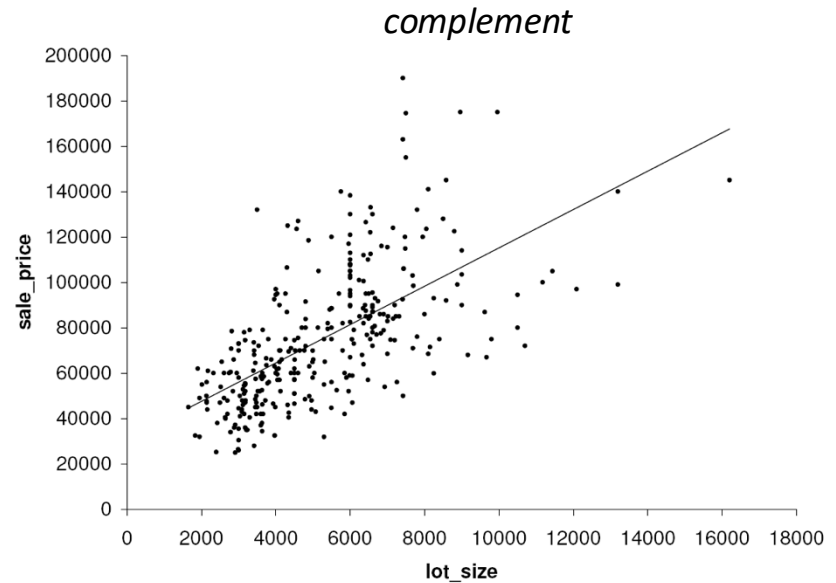
$$y_i = \bar{a} + \bar{b} \cdot x_i, \text{ and}$$

$$y_i = a + b \cdot x_i$$

- Compute significance of slope difference φ_{ssd}

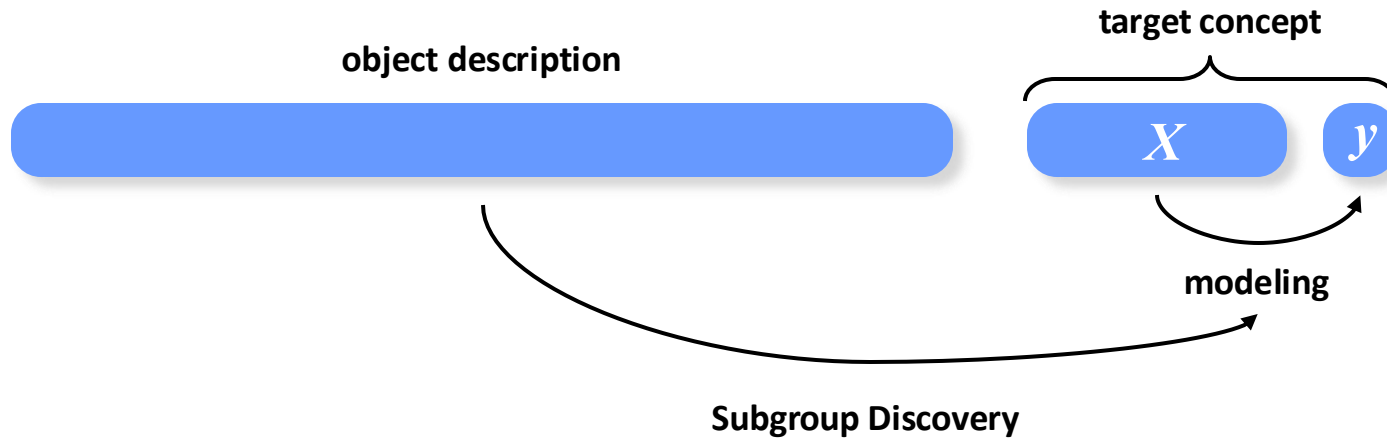


$$y = 41\,568 + 3.31 \cdot x$$



$$y = 30\,723 + 8.45 \cdot x$$

General Framework



General Framework

